

Actividad Integradora

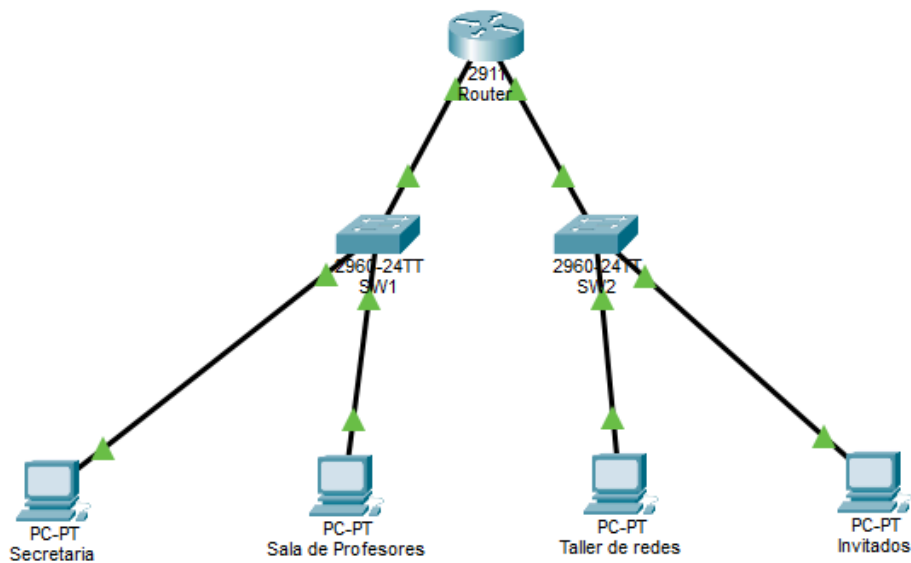


Tabla de Direccionamiento:

PC	Puerto	IPv4	VLAN
Secretaria	SW1 - Fa0/1	192.168.10.10/24	10
Sala de Profesores	SW1 - Fa0/2	192.168.20.10/24	20
Taller de Redes	SW2 - Fa0/1	192.168.30.10/24	30
Sala de Invitados	SW2 - Fa0/2	192.168.40.10/24	40
Router		IPv4	Máscara
Router 2911	SW1 - G0/0.10	192.168.10.1	255.255.255.0
Router 2911	SW1 - G0/0.20	192.168.20.1	255.255.255.0
Router 2911	SW2 - G0/0.30	192.168.30.1	255.255.255.0
Router 2911	SW2 - G0/0.40	192.168.40.1	255.255.255.0

Configuraciones Clave:

Switch1

Configuración de VLANs

```
enable
configure terminal
```

```
vlan 10
name Secretaria
exit
```

```
vlan 20
name Profesores
exit
```

Configuración de PC de secretaria

```
interface range fa0/1
switchport mode access
switchport access vlan 10
exit
```

Configuración de PC de sala de profesores

```
interface range fa0/2
switchport mode access
switchport access vlan 20
exit
```

Configuración trunk para Router

```
interface gigabitEthernet0/1
switchport mode trunk
exit
```

Switch2

Configuración de VLANs

```
vlan 30
name Taller
exit
```

```
vlan 40
name Invitados
exit
```

Configuración de PC de Taller de redes

```
interface range fa0/1
switchport mode access
switchport access vlan 30
exit
```

Configuración de PC para sala de invitados

```
interface range fa0/2
switchport mode access
switchport access vlan 40
exit
```

Configuración trunk para Router

```
interface gigabitEthernet0/1
switchport mode trunk
exit
```

Router

Configuración de VLANs 10, 20, 30 y 40

```
enable
configure terminal
```

```
interface gigabitEthernet0/0
no shutdown
exit
```

```
interface gigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit
```

```
interface gigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
```

```
interface gigabitEthernet0/2.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
exit
```

```
interface gigabitEthernet0/2.40
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
exit
```


	Sala de profesores
ping	<pre> C:\>ping 192.168.20.10 Pinging 192.168.20.10 with 32 bytes of data: Reply from 192.168.20.10: bytes=32 time<1ms TTL=127 Reply from 192.168.20.10: bytes=32 time<1ms TTL=127 Reply from 192.168.20.10: bytes=32 time<1ms TTL=127 Reply from 192.168.20.10: bytes=32 time<1ms TTL=127 Ping statistics for 192.168.20.10: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms </pre>
tracert	<pre> C:\>tracert 192.168.20.10 Tracing route to 192.168.20.10 over a maximum of 30 hops: 0 0 ms 0 ms 0 ms 192.168.10.1 1 1 ms 0 ms 0 ms 192.168.20.10 Trace complete. </pre>
ipconfig	<pre> C:\>ipconfig FastEthernet0 Connection:(default port) Connection-specific DNS Suffix...: Link-local IPv6 Address : FE80::201:42FF:FE75:2EA5 IPv6 Address : :: IPv4 Address. : 192.168.20.10 Subnet Mask : 255.255.255.0 Default Gateway : :: 192.168.20.1 </pre>

	Taller de redes
ping	<pre> C:\>ping 192.168.30.10 Pinging 192.168.30.10 with 32 bytes of data: Reply from 192.168.30.10: bytes=32 time<1ms TTL=127 Reply from 192.168.30.10: bytes=32 time<1ms TTL=127 Reply from 192.168.30.10: bytes=32 time<1ms TTL=127 Reply from 192.168.30.10: bytes=32 time<1ms TTL=127 Ping statistics for 192.168.30.10: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms </pre>

tracert	<pre> C:\>tracert 192.168.30.10 Tracing route to 192.168.30.10 over a maximum of 30 hops: 1 0 ms 0 ms 0 ms 192.168.10.1 2 1 ms 0 ms 0 ms 192.168.30.10 Trace complete. </pre>
ipconfig	<pre> C:\>ipconfig FastEthernet0 Connection:(default port) Connection-specific DNS Suffix...: Link-local IPv6 Address.....: FE80::201:43FF:FEE1:44C1 IPv6 Address.....: :: IPv4 Address.....: 192.168.30.10 Subnet Mask.....: 255.255.255.0 Default Gateway.....: :: 192.168.30.1 </pre>

	Sala de Invitados
ping	<pre> C:\>ping 192.168.40.10 Pinging 192.168.40.10 with 32 bytes of data: Reply from 192.168.40.10: bytes=32 time<1ms TTL=127 Reply from 192.168.40.10: bytes=32 time<1ms TTL=127 Reply from 192.168.40.10: bytes=32 time<1ms TTL=127 Reply from 192.168.40.10: bytes=32 time<1ms TTL=127 Ping statistics for 192.168.40.10: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 0ms, Average = 0ms </pre>
tracert	<pre> C:\>tracert 192.168.40.10 Tracing route to 192.168.40.10 over a maximum of 30 hops: 1 0 ms 0 ms 0 ms 192.168.10.1 2 0 ms 0 ms 0 ms 192.168.40.10 Trace complete. </pre>

ipconfig	<pre> FastEthernet0 Connection: (default port) Connection-specific DNS Suffix...: Link-local IPv6 Address.....: FE80::20B:BEFF:FE60:A961 IPv6 Address.....: :: IPv4 Address.....: 192.168.40.10 Subnet Mask.....: 255.255.255.0 Default Gateway.....: :: 192.168.40.1 </pre>
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No sabemos/No vimos(en este taller)...

5. Justificar y aplicar un criterio de calidad de servicio (QoS): priorizar, por ejemplo, el tráfico del Taller de Redes por sobre el de la Sala de Invitados.
6. Identificar qué dispositivos de la topología requerirían alimentación Power over Ethernet (por ejemplo, el access point) y fundamentar la elección.
7. Configurar el punto de acceso inalámbrico: SSID, cifrado WPA2 o WPA3, contraseña robusta y aislamiento de clientes para la red de invitados.

(También hay una mitad de la clase que sabe lo que es VLAN y/o el puerto trunk porque la primera mitad del año vimos esos temas con el profe de otro taller, mas no fue explicado en este taller y hay compañeros que todavía no saben que es)

DNS

Router

Switch

Contraseña robusta

Aislamiento de clientes